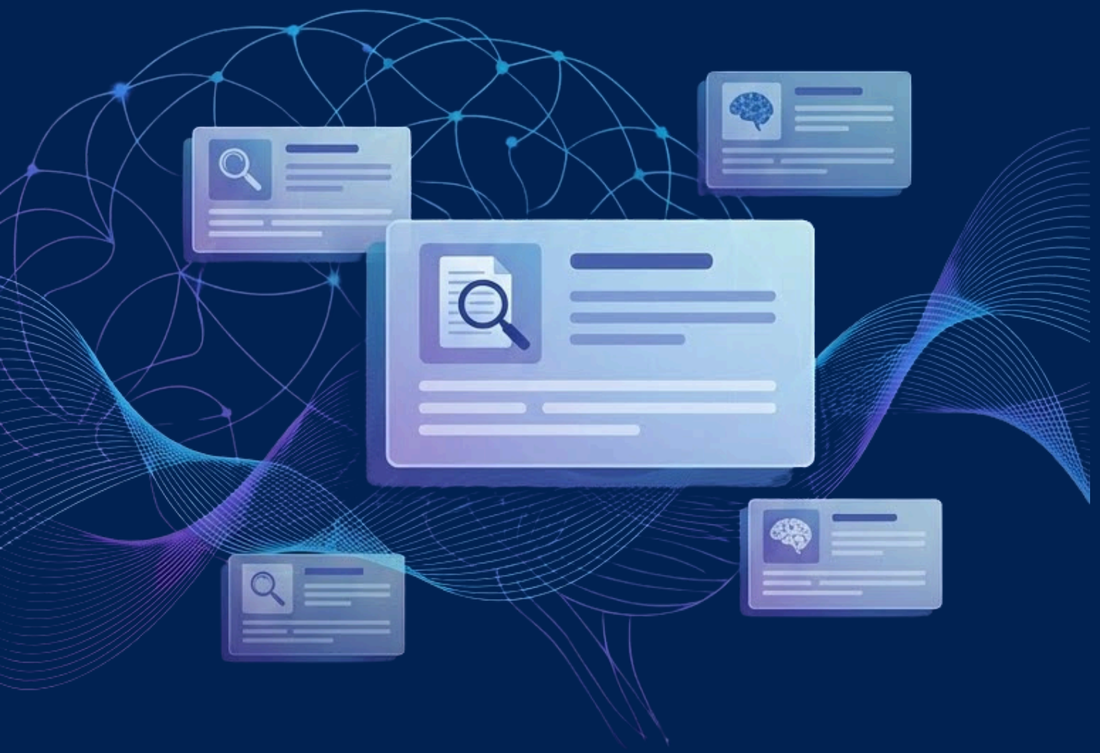


JobCheck – Milestone 2

Detecting Fake Job Posts Using Natural Language Processing

Model Training & Evaluation

INFOSYS SPRINGBOARD

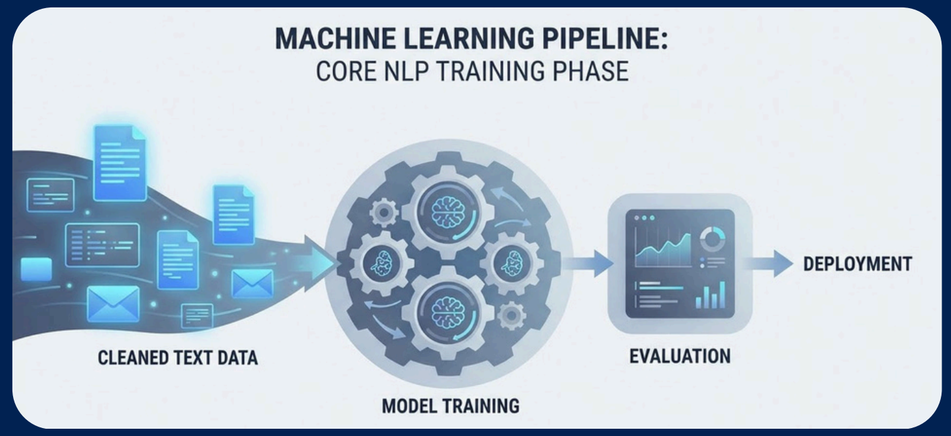
VIRTUAL INTERNSHIP 6.0

Presented by: Mohamed Aslam N

MILESTONE 2

Building Intelligence: The Training Phase

Milestone 2 represents the core machine learning phase of the JobCheck project, where we transform cleaned data into intelligent classification models. This critical stage bridges data preparation and deployment, focusing on building robust models that can accurately distinguish between genuine job opportunities and fraudulent postings.



Our approach combines traditional machine learning techniques with modern deep learning architectures to create a comprehensive detection system.

01

Data Foundation

Leveraging preprocessed data from Milestone 1

02

Model Training

Building classification algorithms

03

Performance Analysis

Evaluating and comparing results

Core Objectives: What We Aim to Achieve

Train Baseline Models

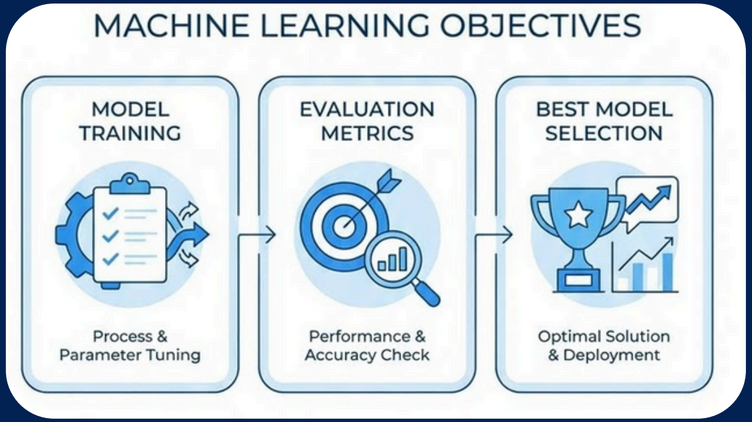
Develop multiple machine learning models using different algorithms to establish performance benchmarks and understand which approaches work best for job post classification.

Rigorous Evaluation

Measure model performance using industry-standard metrics including accuracy, precision, recall, and F1-score to ensure balanced detection capabilities.

Select Best Model

Conduct comprehensive comparison across multiple dimensions—accuracy, speed, and complexity—to identify the optimal model for production deployment.



The Model Arsenal: Our Classification Approaches

1

Logistic Regression

A fast, interpretable linear classifier that serves as our baseline. Despite its simplicity, logistic regression often delivers surprisingly strong performance on text classification tasks when paired with well-engineered features.

- Optimised for speed and interpretability
- Excellent baseline performance

2

Random Forest

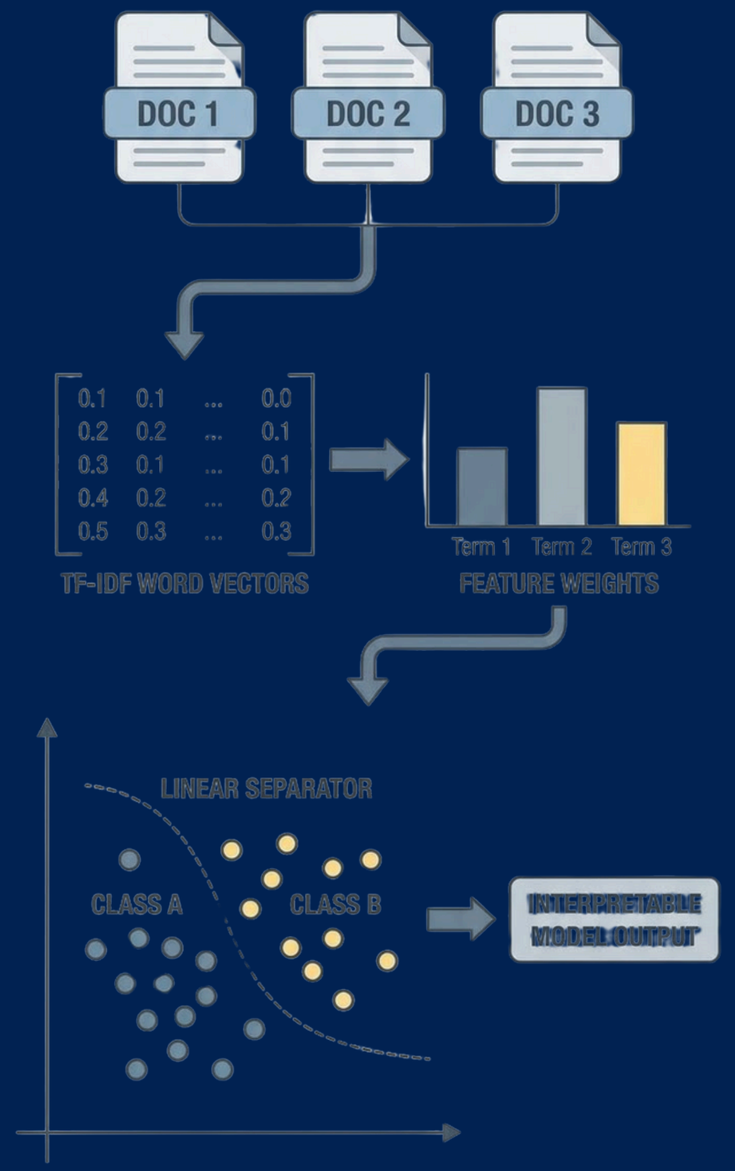
An ensemble learning method that combines multiple decision trees to create robust predictions. Random forests excel at capturing complex patterns and provide valuable insights through feature importance rankings.

- Handles non-linear relationships
- Provides feature importance insights

Both models were selected to establish strong baseline performance before exploring more complex architectures.

MODEL 1

Logistic Regression: Fast & Interpretable



Implementation Details

- **Feature Engineering**

  Trained on TF-IDF vectorised text features, capturing word importance across the job post corpus

- **Hyperparameter Tuning**

  Optimised regularisation strength (C), penalty type (L1/L2), and solver algorithm using grid search

- **Training Success**

  Model converged efficiently on preprocessed data and achieved strong baseline metrics

- **Model Persistence**

  Saved using joblib for efficient reuse and deployment

MODEL 2

Random Forest: Power of Ensembles

Ensemble Architecture

Random Forest combines multiple decision trees, with each tree voting on the final classification. This ensemble approach reduces overfitting and improves generalisation.

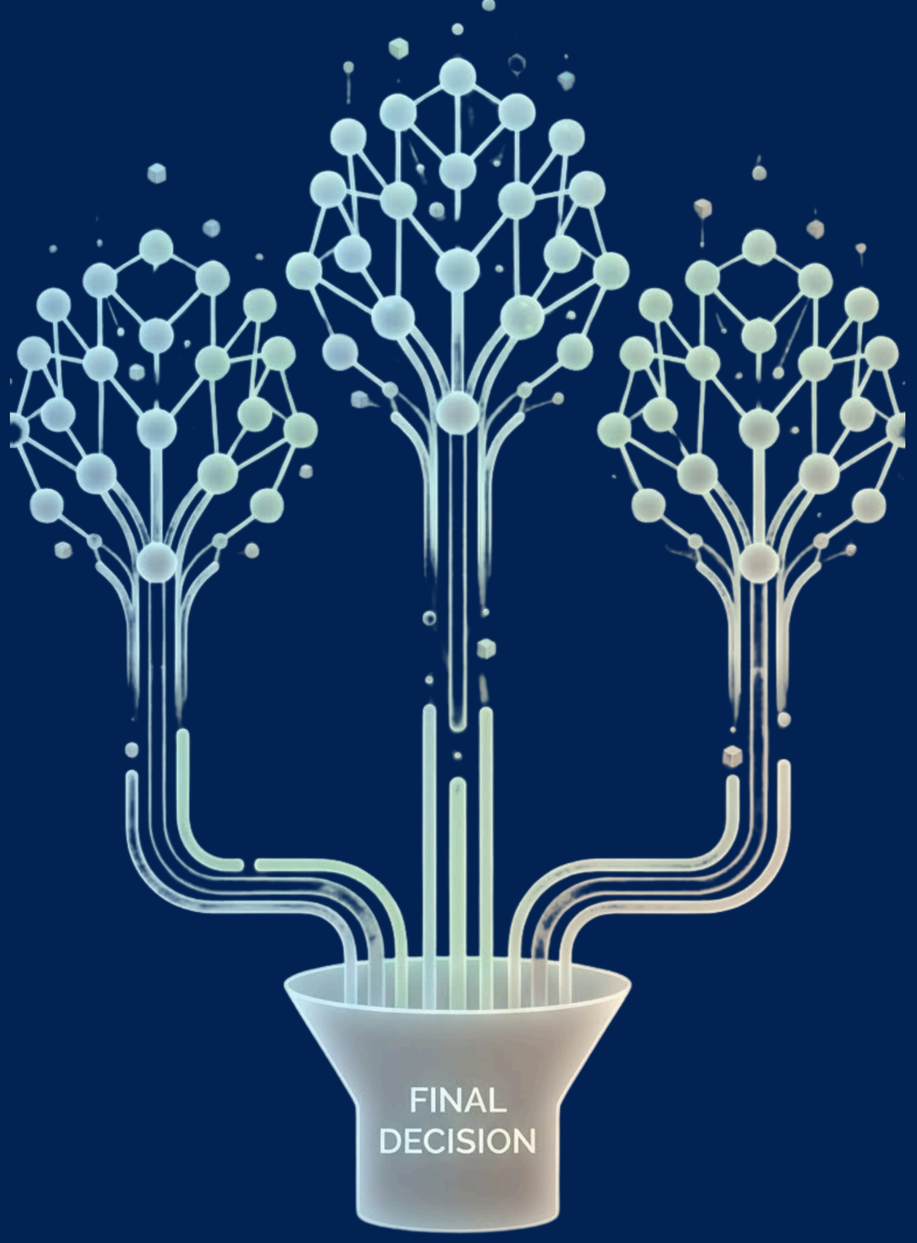
Optimisation Strategy

Key parameters tuned include number of estimators (trees), maximum tree depth, minimum samples per split, and feature sampling strategy for optimal performance.

Feature Insights

Analysed feature importance rankings to understand which words and patterns most strongly indicate fraudulent job postings, providing interpretability.

Model Saved: Random Forest classifier serialised with optimised parameters for comprehensive model comparison and potential production use.



MACHINE LEARNING:
RANDOM FOREST CLASSIFIER

Measuring Success: Our Evaluation Framework

Comprehensive evaluation is crucial for understanding model performance beyond simple accuracy. We employ multiple metrics to assess different aspects of classification quality, ensuring our model performs well for both real and fake job post detection.

Accuracy

Overall correctness: proportion of all predictions that were classified correctly

Precision

Of all posts flagged as fake, what percentage were actually fraudulent?

Recall

Of all actual fake posts, what percentage did our model successfully identify?

F1-Score

Harmonic mean of precision and recall, providing a balanced performance measure

Confusion Matrix

Visual breakdown of true positives, true negatives, false positives, and false negatives

ROC-AUC Score

Measures the model's ability to distinguish between classes across all thresholds

ADVANCED APPROACH

Deep Learning: Next-Level Classification

Neural Architecture

To push beyond traditional ML boundaries, we implemented a sophisticated deep learning model capable of learning complex linguistic patterns and contextual relationships within job post text.

1

Embedding Layer

Word2Vec, GloVe, or BERT embeddings capture semantic meaning

2

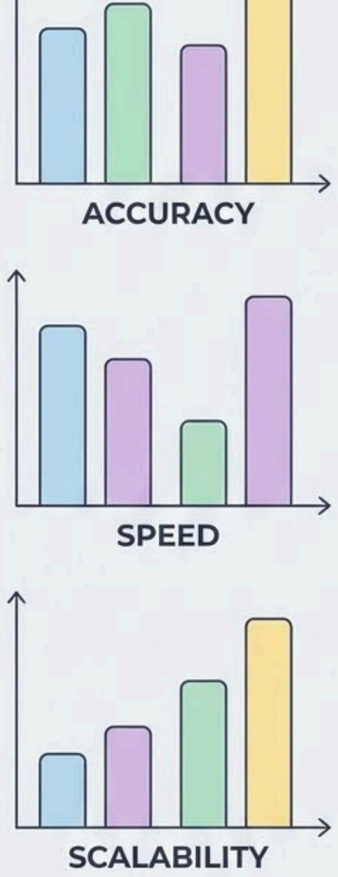
BiLSTM Architecture

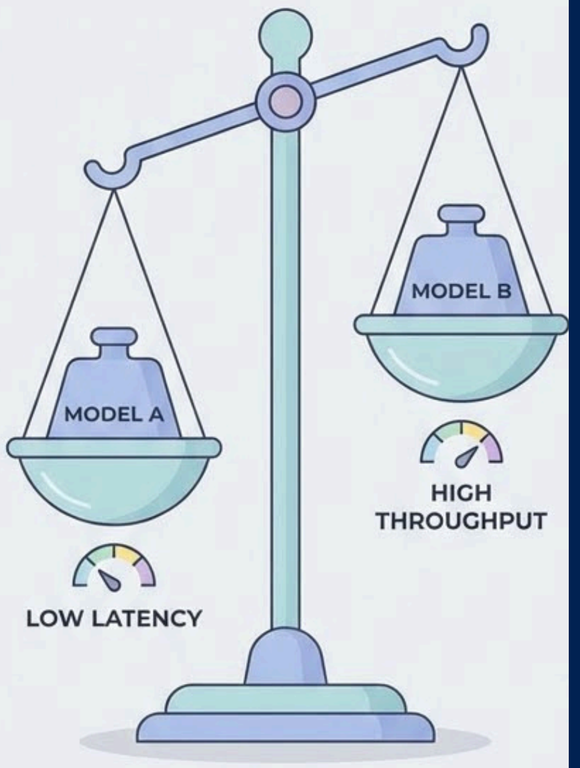
Bidirectional processing captures context from both directions

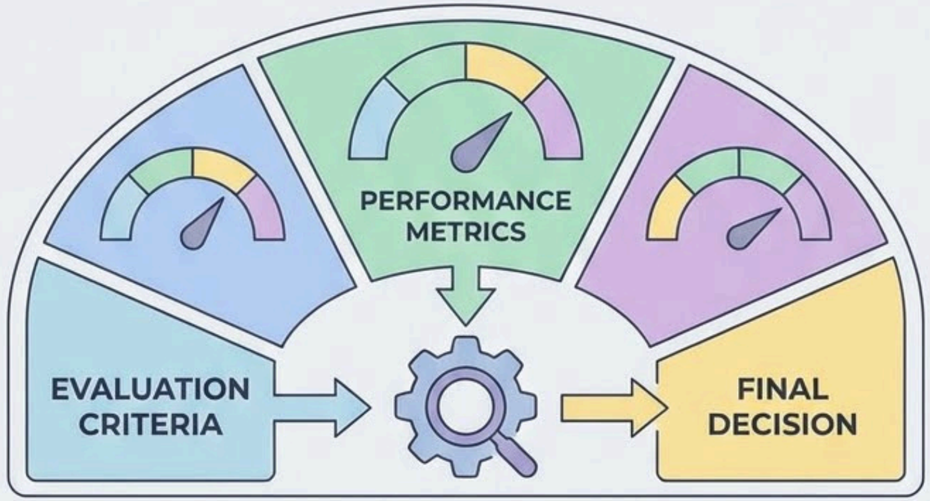
3

Output Classification

Dense layers with sigmoid activation for binary prediction







Model Showdown: Selecting Our Champion

Comprehensive Comparison Framework

Choosing the optimal model requires evaluating multiple dimensions beyond raw accuracy. We conducted side-by-side analysis of all trained models, considering real-world deployment constraints and business requirements.

Performance Metrics

Accuracy, precision, recall, F1-score, and ROC-AUC compared across all models

Inference Speed

Prediction latency measured to ensure real-time classification capability

Model Complexity

Memory footprint, training time, and interpretability assessed for production viability

Generalisation

Performance on unseen data evaluated through cross-validation and holdout sets

Best Model Selected: Based on balanced performance across all criteria. Model versioning implemented using MLflow for tracking experiments and reproducibility.

Journey Insights & Forward Path

Key Learnings from Milestone 2

Feature Quality Matters

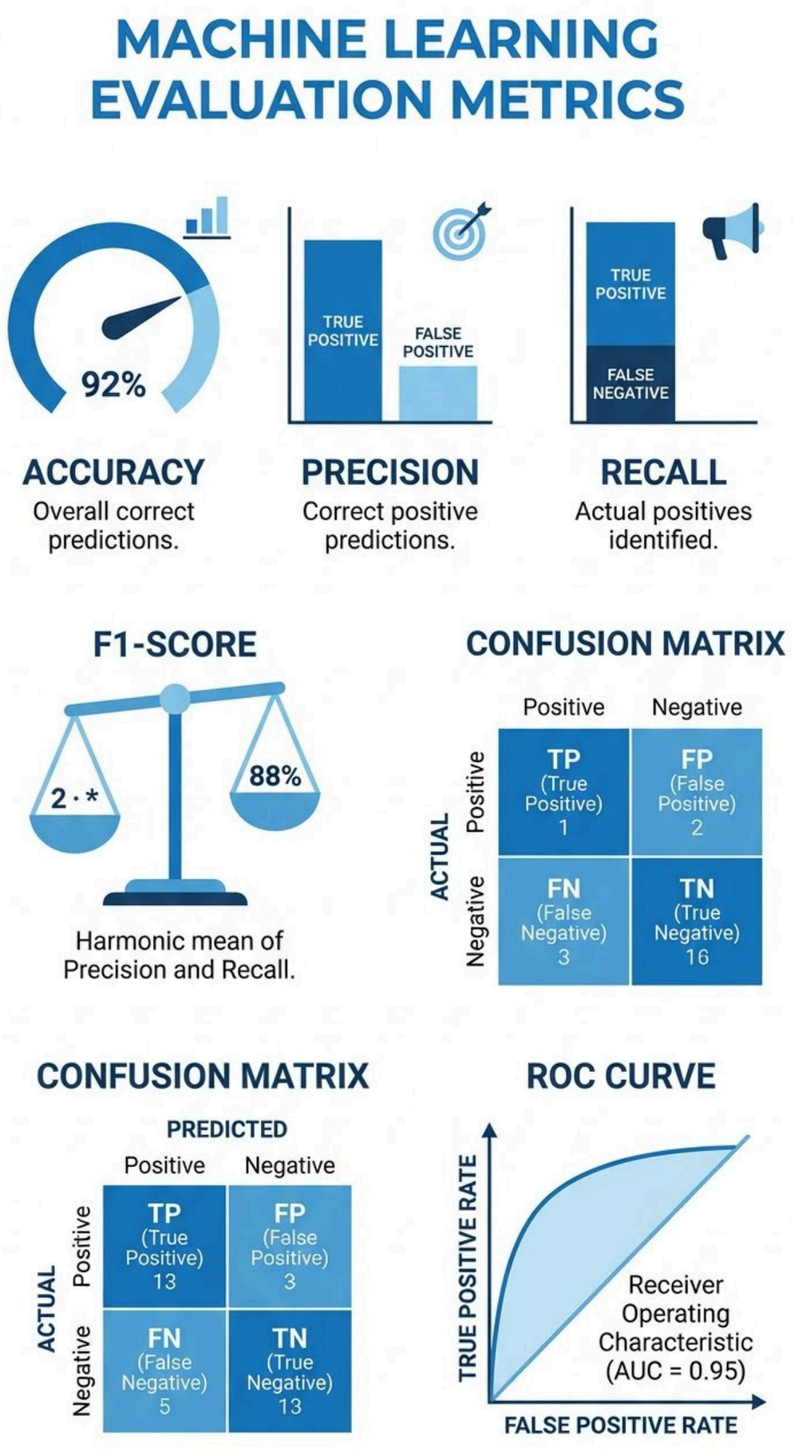
High-quality feature engineering from Milestone 1 directly translated to superior model performance across all algorithms tested.

Multiple Models Required

Testing diverse algorithms revealed significant performance variations, validating the importance of comprehensive model exploration.

Balanced Metrics Critical

Relying solely on accuracy would have been misleading; precision-recall balance ensures real-world reliability.

Iterative Refinement

Model improvement is an ongoing process requiring continuous evaluation and hyperparameter optimisation.